Autonomous Research Agent (ARA)

Orchestrating AI for Q1-Grade Literature Reviews

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Vision & Problem

The Vision of ARA

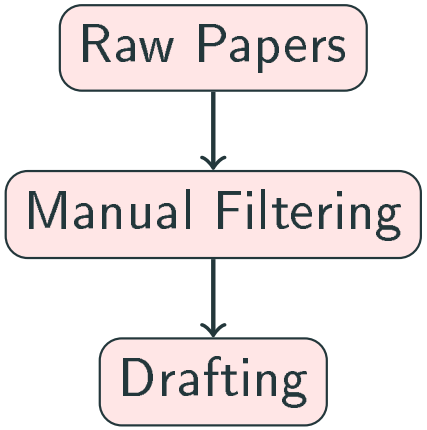
- **Autonomous Mastery**: Transform a simple research theme into a 30-page academic manuscript.
- **High-Impact Focused**: Targeting Q1 journal standards through rigorous bibliometric analysis.
- **Agentic Intelligence**: Moving beyond templates to dynamic, self-planning synthesis.

The Goal

Bridging the gap between raw web data and structured, peer-reviewable academic knowledge.

The Problem: Information Overload

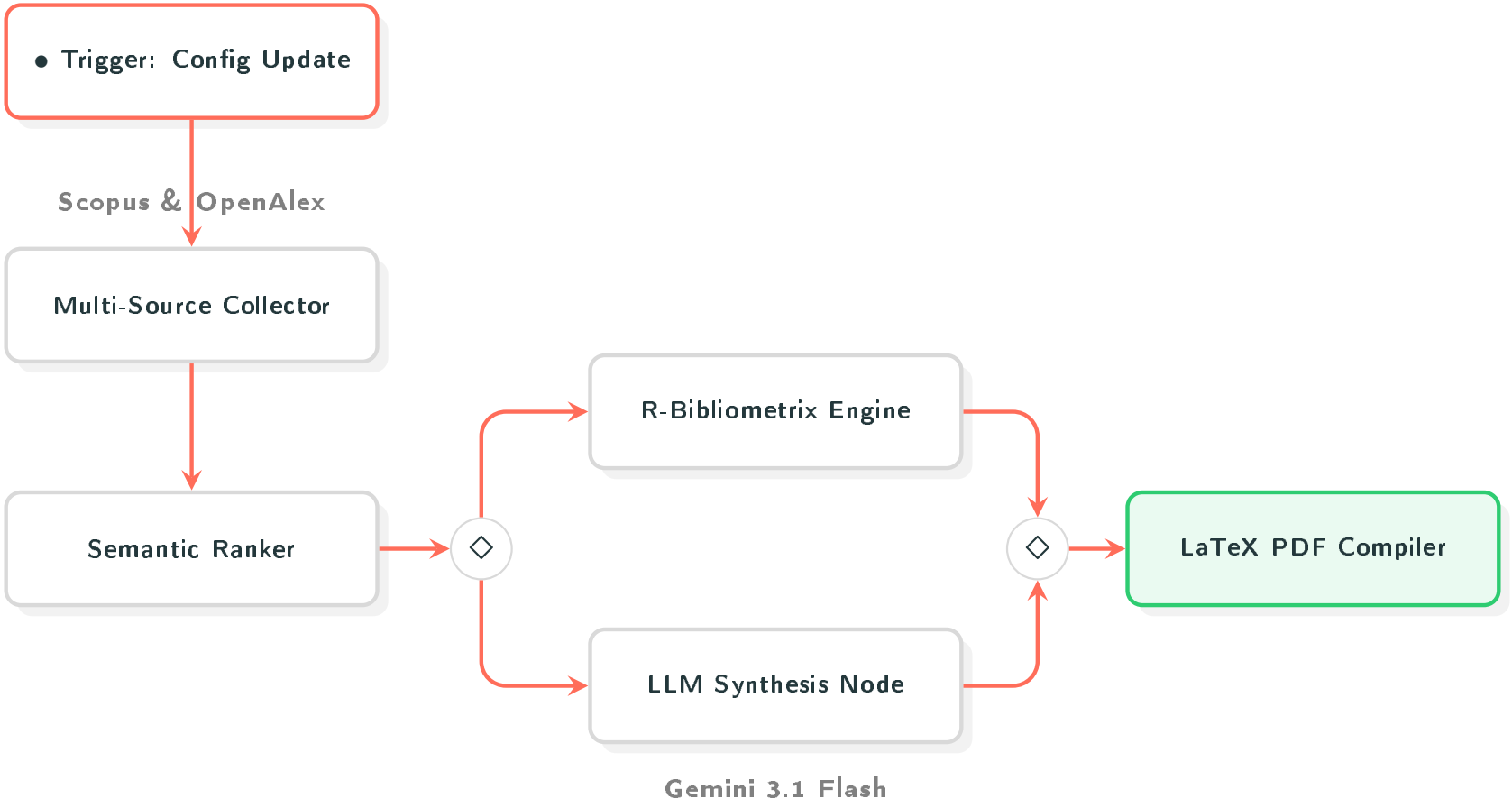
- 5M+ papers published annually.
- "Hallucination" risks in standard LLMs.
- Disconnected data: Bibliometry vs. Synthesis.



Weeks of work

Global Workflow

Global Workflow Orchestration



A modular execution graph where each node represents a specific agentic responsibility.

Brainstorming: Why a Modular Workflow?

Design Strategy
We use a **Pipeline** pattern rather than a single LLM call.

Decoupling : If Scopus API changes, only one module needs repair.

Groundedness : The LLM "Drafter" only uses papers "Ranked" by the system—Zero Hallucination.

Diversity : Combining Python (data), R (stats), and LaTeX (design).

Functional Deep Dive

Node 1: Multi-Source Intelligent Collector

Functionality: Hybrid scholarly retrieval (Scopus + OpenAlex).

- **Dual Integration**: Supports Scopus (Elsevier) and OpenAlex (Open Access).
- **Deduplication Engine**: Merges results from multiple sources and removes duplicates based on DOI and Title.
- **AI Query Engine**: Uses Gemini to generate optimized search queries for each database.
- **Abstract Reconstruction**: Resolves OpenAlex's inverted index into readable text.

Node 2: Semantic Paper Ranker

Functionality: Sifting the gold from the noise.

- **Scoring Formula**: $Score = \alpha \cdot \text{Citations} + \beta \cdot \text{Recency}$.
- **Primary Input**: Provides the "Top 15" articles to the LLM.
- **JSON Link**: Outputs *ranked_articles.json* for downstream orchestration.

Node 3: Bibliometric Engine (R)

Functionality: Science Mapping.

- **Integration**: Python $\rightarrow$ R scripting bridge.
- **Key Visuals**: Thematic Maps, Co-citation Networks, and Authors' Production over time.
- **Standard**: Uses the internationally recognized 'bibliometrix' R package.

Node 4: Dynamic LLM Drafter

Functionality: Content Synthesis.

- **Dynamic Planning**: Gemini analyzes articles to propose a custom Table of Contents.
- **Section-by-Section Synthesis**: High-density academic drafting (3000+ words).
- **LaTeX Native**: Content is pre-escaped and citation-aware (‘idX‘).

Orchestration & Tuning

Parameter Spotlight: agent_config.json

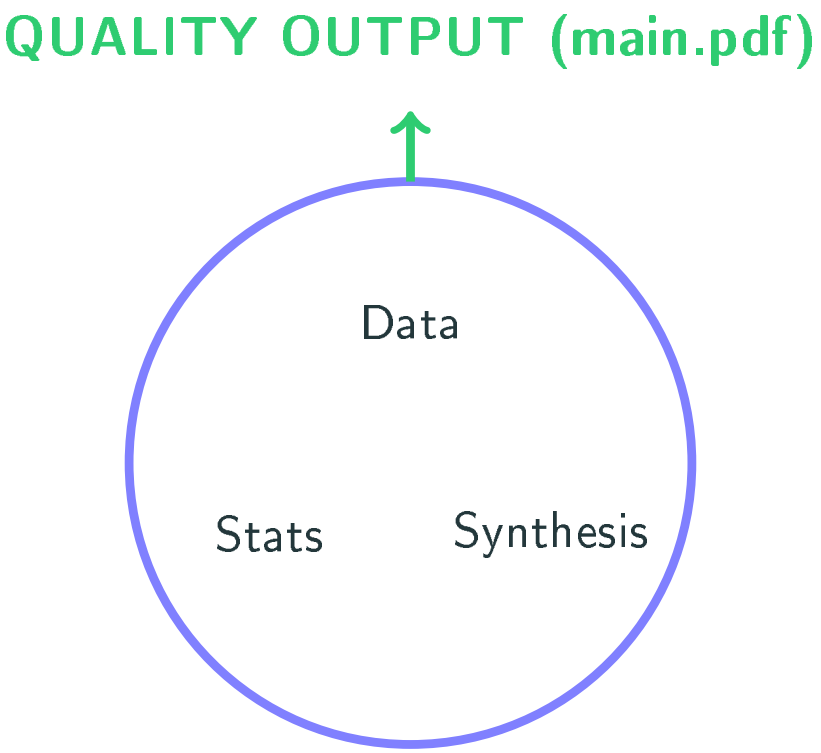
```
{
  "scopus_api_key": "...",
  "openalex_api_key": "...",
  "article_count": 50,
  "intro_word_count": 300,
  "review_word_count": 3000,
  "theme_description": "...",
  "model_name": "gemini-3.1-flash-lite-preview"
}
```

- **Granularity**: Control depth vs. breadth.
- **Hybrid Collection**: Toggle between Scopus, OpenAlex, or both.

Resilience: Handling Quotas (429 Errors)

- **Wait Strategy**: Automatic cooldown (10s) between API hits.
- **Retry Logic**: Up to 5 retries with increasing delay.
- **Context Compression**: Truncating abstracts to save on Free-Tier TPM limits.

System Convergence



Results & Impact

Case Study: Decarbonization & Equity

- **Input**: Theme on intergenerational fiscal structures.
- **Process**: 25 Scopus papers retrieved, AI generated 5 thematic sections.
- **Output**: Comprehensive 12-page PDF with 35+ citations, fully compiled.

Critique & Recommendations

Current System Limitations

- **API Quotas**: Still bound by Gemini and Elsevier/OpenAlex rate limits.
- **Linear Workflow**: Progress is sequential; lacks parallel multi-agent negotiation.
- **Human-in-the-loop**: Minimal intervention for real-time outline editing.
- **Computational Overhead**: R-Bibliometrix requires a full local R environment.

Recommendations for Future Evolution

- **Cloud Parallelism**: Offload R analytics to containerized serverless functions.
- **Hybrid LLM Core**: Use an ensemble of models (e.g., Gemini for speed, Claude for creative nuance).
- **Web Orchestrator**: Deploy as a FastAPI backend with a React visual dashboard.
- **Zotero/Mendeley Bridge**: Automated export to reference management software.

Future Horizons

- **Knowledge Graphs**: Visualizing citation "trees" directly in LaTeX.
- **Real-time Synthesis**: Streaming generation of sections for interactive research.

Autonomous Research is No Longer Science Fiction.

The ARA project demonstrates that by combining modular architecture, R-based analytics, and LLM-driven synthesis, we can accelerate scientific discovery by orders of magnitude.

Questions? Contact: Bilel Ammouri